

People Analytics

Groupwork Use Cases 4 – Predicting Employee Performance

In this workshop, you will explore multiple models to predict individual, team and organizational performance. You will apply multiple regression, stepwise regression, and dummy variable techniques to performance data from various sectors. Your goal is to derive actionable insights about what drives performance and how organizations can use these insights ethically and strategically.

Prompt 1: Predicting Customer Loyalty (Financial Services)

Dataset: customer_satisfaction.xlsx

Tasks:

- Estimate a multiple regression model with Custloyalty as the dependent variable. Predictors: Sat1–Sat4 and SexOfSalesperson.
- Interpret R^2 , F-statistic and beta coefficients.
- Recommend areas for sales improvement.

Deliverables:

- Model output and standardized coefficients.
- Strategic recommendation for salesperson development.

Prompt 2: Predicting Customer Reinvestment (Financial Services)

Dataset: customer_satisfaction.xlsx

Tasks:

- Replicate Prompt 1 using InvestMore as the dependent variable.
- Compare results and R^2 .
- Discuss implications for upselling and cross-selling strategy.

Deliverables:

- Model summary and interpretation.
- Recommendations on customer retention strategies.

Prompt 3: Linking Employee Engagement to Customer Loyalty

Dataset: aggregate_sales_customer_loyalty.xlsx

Tasks:

- Predict average customer loyalty per salesperson. Predictors: qEng1, qEng2, qMan1, Talent2014, Sex, Age, Manager.
- Evaluate significance and limitations.
- Discuss engagement-performance links.

Deliverables:

- Regression output and interpretation.
- Organizational insight and caveats.

Prompt 4: Predicting Individual Performance Ratings (Manufacturing)

Dataset: perf_2014_to_2015_sick_2014.xlsx

Tasks:

- Predict 2015 performance using 2014 predictors. Variables: POS2014, DistJust2014, ProcJust2014, JOBSAT2014, ValueFit2014, JobStrain2014, SickDays2014.
- Interpret R^2 and beta values.
- Discuss what drives performance improvements.

Deliverables:

- Model summary and insights.
- HR recommendations based on findings.

Prompt 5: Stepwise Regression for Performance Prediction

Dataset: perf_2014_to_2015_sick_2014.xlsx

Tasks:

- Use stepwise regression to select significant predictors.
- Compare model fit with Prompt 4.
- Evaluate parsimony and implications for policy.

Deliverables:

- Stepwise model output.
- Strategic use of simpler models in HR.

Prompt 6: Controlling for Prior Performance Ratings

Dataset: perf_2014_to_2015_sick_2014.xlsx

Tasks:

- Include 2014 performance rating as a predictor.
- Identify whether other variables still predict 2015 performance.
- Explain improvement prediction logic.

Deliverables:

- Output with adjusted R^2 and key drivers.
- Change-focused HR insight.

Prompt 7: Predicting Sickness Absence

Dataset: perf_2014_to_2015_sick_2014.xlsx

Tasks:

- Switch dependent variable to SickDays2014.
- Use attitude variables as predictors.
- Analyze non-significant results and discuss potential.

Deliverables:

- Regression output.
- Interpretation and potential data limitations.

Prompt 8: Exploring Performance Patterns by Demographics

Dataset: MI_data_perf.xlsx

Tasks:

- Predict performance rating using SalaryBenchmark, Gender, AgeCat, TenureCat and Country dummies.
- Highlight impact of gender and country.
- Identify signs of bias in appraisal systems.

Deliverables:

- Regression output.
- Recommendations to improve fairness in performance evaluation.

Prompt 9: Predicting Checkout Performance by Demographics

Dataset: supermarket_checkout.xlsx

Tasks:

- Predict ScanPminute using Gender, education, tenure, store, and age.
- Run model with age dummies to improve fit.
- Discuss age-related productivity insights.
- Address ethical concerns in applying the findings.

Deliverables:

- Two models (with and without dummies).
- Strategic and ethical recommendations for workforce planning.